



School of Computing and Engineering Sciences

BICS

Advanced Database Systems

Data warehouse Assignment

Area: Data Warehousing, Extraction Transformation & Loading and Data Visualization	Tools: Talend Studio for Data Intergration, Postres/MySQL/Oracle Database, Tableau, Any scripting language (Preferably Python, JavaScript or PHP)
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Narrative

Kiva is an international nonprofit, founded in 2005 and based in San Francisco, with a mission to connect people through lending to alleviate poverty. We celebrate and support people looking to create a better future for themselves, their families, and their communities. By lending as little as \$25 on Kiva, anyone can help a borrower start or grow a business, go to school, access clean energy, or realize their potential. For some, it's a matter of survival, for others it's the fuel for a lifelong.

The life cycle of a KIVA loan is as follows: A borrower applies for a loan, the loan goes through the underwriting and approval process, then the load undergoes Loan disbursal period (i.e. when the borrower can access the money— the timing of this can vary), after which the borrower repays the loan and finally Lenders use repayments to fund new loans, donate or withdraw the money.

Project Instructions

You have been approached by the board of the KIVA loan to come up with a modern data warehouse. The board seeks to finally have detailed analytics on the following areas:

- i. The borrowing rate across different times of the year
- ii. The borrowing trend across different regions
- iii. The type of loans mostly applied for
- iv. The disbursement rate across the year
- v. The fundraising trend across different regions in comparison to the time of the year

- vi. The repayment trend in the different regions with the passing of time
- vii. Reports of expenses incurred in the course of different KIVA loan cycles
- viii. The demographic trend of the borrower across different regions in relation to time of the year
- ix. The demographic trend of the fundraisers across different regions in relation to time of the year

All that has been provided for you are four datasets which are attached with this document. You are expected to study the datasets and develop different data marts for each group. If need be, where the dataset is not sufficient, you are expected to come up with additional arbitrary but meaningful datasets. Ensure you do not repeat a dimension or fact from other groups. Remember a data mart consists of dimensions and facts that presuppose a common business logic.

The allocation of reports per group are as follows:

Groups	Reports to generate(From the above list)
1, 5, 7, 13,17	i, iv & ix
2, 6, 8, 14	ii & v
3, 9, 11,15	iii & vi
4, 10, 12,16	viii & vii

Project Deliverables

- 1) Identification of arbitrary dimensions and facts
- 2) Design of star schemas for each question